

InventoryServiceApplicationTests

```
package com.smartlogix.inventory;
```

```
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;
import org.junit.jupiter.api.extension.ExtendWith;
import org.mockito.junit.jupiter.MockitoExtension;
import java.util.HashMap;
import java.util.Map;
```

```
@ExtendWith(MockitoExtension.class)
```

```
public class InventoryServiceApplicationTests {
```

```
    @Test
```

```
    public void testVerificacionStock_DebeCambiarAEstadoQuiebreCuandoEsCero() {
        int cantidadActualEnBodega = 0;
```

```
        Map<String, Object> productoMock = new HashMap<>();
        productoMock.put("sku", "PROD-10023");
        productoMock.put("stock", cantidadActualEnBodega);
```

```
        String estadoResultante = "STOCK_OK";
        int stockEval = (int) productoMock.get("stock");
```

```
        if (stockEval <= 0) {
            estadoResultante = "Quiebre Stock";
        } else if (stockEval <= 5) {
            estadoResultante = "Stock Critico";
        }
    }
```

```
        assertNotNull(productoMock);
        assertEquals("Quiebre Stock", estadoResultante);
    }
```

```
}
```

OrderServiceApplicationTests

```
package com.smartlogix.order;
```

```
import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;
import org.junit.jupiter.api.extension.ExtendWith;
import org.mockito.junit.jupiter.MockitoExtension;
import java.util.HashMap;
import java.util.Map;
```

```
@ExtendWith(MockitoExtension.class)
```

```
public class OrderServiceApplicationTests {
```

```

@Test
public void testValidacionCupon_DebeAplicarDiezPorCientoDescuento() {
    // 1. GIVEN: Datos base de la orden de compra
    String codigoCupon = "DESC10-BIENVENIDA";
    double montoOriginal = 100000;

    // Simulación analítica de la base de datos (H2 / JPA)
    Map<String, Object> cuponMock = new HashMap<>();
    cuponMock.put("code", "DESC10-BIENVENIDA");
    cuponMock.put("active", true);
    cuponMock.put("discountPercentage", 0.10);

    // 2. WHEN: Lógica de negocio del Avance 2
    double montoFinal = montoOriginal;
    boolean esValido = (boolean) cuponMock.get("active") &&
        cuponMock.get("code").equals(codigoCupon);

    if (esValido) {
        double porcentaje = (double) cuponMock.get("discountPercentage");
        double descuento = montoOriginal * porcentaje;
        montoFinal = montoOriginal - descuento;
    }

    // 3. THEN: Aserciones de JUnit 5
    double valorEsperado = 90000;
    assertNotNull(cuponMock);
    assertTrue(esValido);
    assertEquals(valorEsperado, montoFinal, 0.001);
}
}

ShipmentServiceApplicationTests
package com.smartlogix.shipment;

import static org.junit.jupiter.api.Assertions.*;
import org.junit.jupiter.api.Test;
import org.junit.jupiter.api.extension.ExtendWith;
import org.mockito.junit.jupiter.MockitoExtension;
import java.util.HashMap;
import java.util.Map;

@ExtendWith(MockitoExtension.class)
public class ShipmentServiceApplicationTests {

    //
    =====
    ===

```

```

// TEST AVANCE 1: ADAPTADOR MULTICARRIER POR REGIÓN
//
=====
===
@Test
public void testAdaptadorRuta_DebeElegirStarkenParaValparaisoPorPrecio() {
    // 1. GIVEN: Definimos una dirección física perteneciente a la V Región (Destino)
    String destinoValpo = "Calle Limache 1420, Viña del Mar";

    // 2. WHEN: Simulamos el comportamiento lógico exacto del Adaptador Multicarrier
    // Usamos una estructura Map estándar para simular la respuesta JSON de la API
    REST
    Map<String, String> responseMock = new HashMap<>();

    if (destinoValpo.toLowerCase().contains("viña") ||
destinoValpo.toLowerCase().contains("valparaiso")) {
        responseMock.put("carrierAsignado", "Starken");
        responseMock.put("routeCode", "RTE-VALPO-ZONA5");
        responseMock.put("status", "OPTIMIZED_BY_PRICE");
        responseMock.put("costo", "5200");
    } else {
        responseMock.put("carrierAsignado", "Chilexpress");
        responseMock.put("routeCode", "RTE-STGO-DEFAULT");
        responseMock.put("status", "STANDARD");
        responseMock.put("costo", "7500");
    }

    // 3. THEN: Verificaciones unitarias rigurosas exigidas por JUnit 5
    assertNotNull(responseMock, "La respuesta del adaptador no puede ser nula.");

    // Validamos que el algoritmo seleccione Starken por precio para la región de
    Valparaíso
    assertEquals("Starken", responseMock.get("carrierAsignado"),
        "ERROR: Para Viña del Mar el adaptador debió conmutar a Starken por economía.");

    // Validamos que asigne la ruta logística adecuada
    assertEquals("RTE-VALPO-ZONA5", responseMock.get("routeCode"),
        "ERROR: Código de ruta regional incorrecto.");

    // Validamos que el estado de la persistencia sea el optimizado
    assertEquals("OPTIMIZED_BY_PRICE", responseMock.get("status"));
}
}

```

```

PS C:\Users\mr_vi\Downloads\Full_evaluacion2-main> cd
C:\Users\mr_vi\Downloads\Full_evaluacion2-main\Full_evaluacion2-main\inventory-service
>> ..\mvnw test

```

```
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.smartlogix:inventory-service >-----
[INFO] Building inventory-service 1.0.0
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ inventory-service ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ inventory-service ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ inventory-service ---
[INFO] skip non existing resourceDirectory
C:\Users\mr_vi\Downloads\Full_evaluacion2-main\Full_evaluacion2-main\inventory-service\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ inventory-service ---
[INFO] Recompiling the module because of added or removed source files.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to
target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ inventory-service ---
[INFO] Using auto detected provider
org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.smartlogix.inventory.InventoryServiceApplicationTests
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes
because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.974 s -- in
com.smartlogix.inventory.InventoryServiceApplicationTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.874 s
[INFO] Finished at: 2026-07-01T14:23:24-04:00
[INFO] -----
```

PS

C:\Users\mr_vi\Downloads\Full_evaluacion2-main\Full_evaluacion2-main\inventory-service>

cd C:\Users\mr_vi\Downloads\Full_evaluacion2-main\Full_evaluacion2-main\order-service

>> ..\mvnw test

[INFO] Scanning for projects...

[INFO]

[INFO] -----< com.smartlogix:order-service >-----

[INFO] Building order-service 1.0.0

[INFO] from pom.xml

[INFO] -----[jar]-----

[INFO]

[INFO] --- resources:3.4.0:resources (default-resources) @ order-service ---

[INFO] Copying 1 resource from src\main\resources to target\classes

[INFO]

[INFO] --- compiler:3.13.0:compile (default-compile) @ order-service ---

[INFO] Nothing to compile - all classes are up to date.

[INFO]

[INFO] --- resources:3.4.0:testResources (default-testResources) @ order-service ---

[INFO] skip non existing resourceDirectory

C:\Users\mr_vi\Downloads\Full_evaluacion2-main\Full_evaluacion2-main\order-service\src\test\resources

[INFO]

[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ order-service ---

[INFO] Recompiling the module because of changed source code.

[INFO] Compiling 1 source file with javac [debug parameters release 17] to target\test-classes

[INFO]

[INFO] --- surefire:3.5.4:test (default-test) @ order-service ---

[INFO] Using auto detected provider

org.apache.maven.surefire.junitplatform.JUnitPlatformProvider

[INFO]

[INFO] -----

[INFO] T E S T S

[INFO] -----

[INFO] Running com.smartlogix.order.OrderServiceApplicationTests

OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended

[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.846 s -- in com.smartlogix.order.OrderServiceApplicationTests

[INFO]

[INFO] Results:

[INFO]

[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

[INFO]

[INFO] -----

[INFO] BUILD SUCCESS

[INFO] -----

[INFO] Total time: 5.554 s

```
[INFO] Finished at: 2026-07-01T14:23:50-04:00
[INFO] -----
PS C:\Users\mr_vi\Downloads\Full_evaluacion2-main\Full_evaluacion2-main\order-service>
cd
C:\Users\mr_vi\Downloads\Full_evaluacion2-main\Full_evaluacion2-main\shipment-service
>> ..\mvnw test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.smartlogix:order-service >-----
[INFO] Building order-service 1.0.0
[INFO]   from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.4.0:resources (default-resources) @ order-service ---
[INFO] Copying 1 resource from src\main\resources to target\classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ order-service ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.4.0:testResources (default-testResources) @ order-service ---
[INFO] skip non existing resourceDirectory
C:\Users\mr_vi\Downloads\Full_evaluacion2-main\Full_evaluacion2-main\shipment-service\s
rc\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ order-service ---
[INFO] Recompiling the module because of added or removed source files.
[INFO] Compiling 1 source file with javac [debug parameters release 17] to
target\test-classes
[INFO]
[INFO] --- surefire:3.5.4:test (default-test) @ order-service ---
[INFO] Using auto detected provider
org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.smartlogix.shipment.ShipmentServiceApplicationTests
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes
because bootstrap classpath has been appended
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.765 s -- in
com.smartlogix.shipment.ShipmentServiceApplicationTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
```

[INFO] -----

[INFO] Total time: 4.750 s

[INFO] Finished at: 2026-07-01T14:24:05-04:00

[INFO] -----

PS

C:\Users\mr_vi\Downloads\Full_evaluacion2-main\Full_evaluacion2-main\shipment-service>